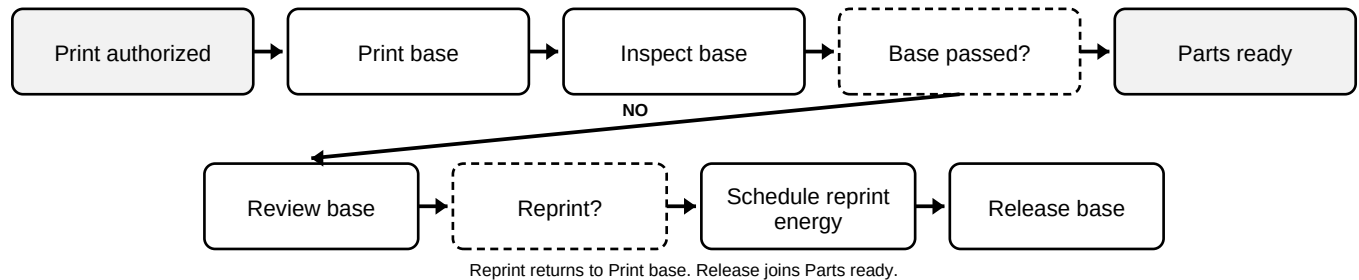


BPMN ACTIVITIES AT THIS TABLE

- Print base
- Inspect base
- Review base
- Schedule reprint energy
- Release base

PROCESS FRAGMENT



HARDWARE AND EVENT CAPTURE

 P2S local telemetry VERIFY	Print start/end, job ID, fault
 Plug energy meter NEEDED	Power and cumulative job energy
 MLX90640 thermal array VERIFY	Heating and cooling episode
 XIAO or IMX335 camera VERIFY	Base present and removed
 Inspection button or UI CONFIGURE	Quality decision and release/reprint
 Raspberry Pi edge node ASSIGN	Source time, adapter health, buffering

START AT THIS TABLE

- 1 Clear the printer and confirm ventilation and fire controls.
- 2 Load the approved design revision and material.
- 3 Connect the edge node and verify the printer, meter, thermal array, and camera signals.
- 4 Check the camera region and zero the energy baseline.
- 5 Bind the work order, schedule authorization, configuration, and printer job ID.
- 6 Start capture before the print. Record inspection and release or reprint after cooling.

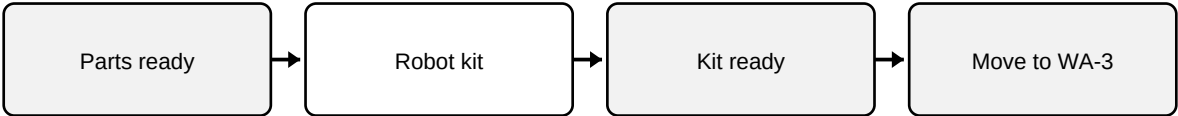
COMPLETE WHEN THIS EVIDENCE EXISTS

Authorization ID, print-job ID, component ID, measured energy, printer result, inspection decision, and evidence timestamps.

BPMN ACTIVITIES AT THIS TABLE

- Robot kit

PROCESS FRAGMENT



Controller success and expected bin changes are both required.

HARDWARE AND EVENT CAPTURE

 PAROL6 controller VERIFY	Robot-job start/end and fault
 Instrumented bin load cells CONFIGURE	Expected parts removed and kit ready
 IMX335 or XIAO camera VERIFY	Part and prepared-kit presence
 Guard and E-stop input NEEDED	Safety state only; never a research override
 Raspberry Pi edge node ASSIGN	Controller adapter, clock, local buffer

START AT THIS TABLE

- 1 Validate the independent guard and emergency stop before robot power-up.
- 2 Clear the reach area. Home the robot with no person inside the guarded area.
- 3 Load the correct electronics into labeled bins and tare the load cells.
- 4 Connect the controller adapter and verify the camera region.
- 5 Bind the work order, robot-job ID, and expected kit contents.
- 6 Start capture before motion. Complete only after controller success and matching bin changes.

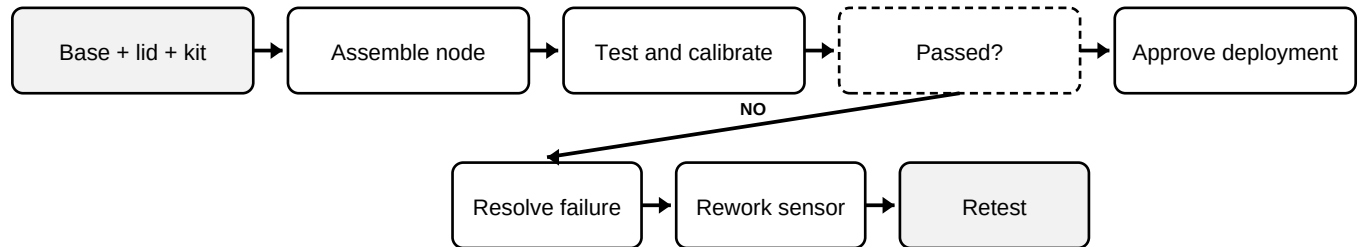
COMPLETE WHEN THIS EVIDENCE EXISTS

Robot-job ID, controller lifecycle, fault state, expected bin deltas, kit identity, and workpiece-presence evidence.

BPMN ACTIVITIES AT THIS TABLE

- Assemble node
- Test and calibrate
- Resolve failure
- Rework sensor
- Approve deployment

PROCESS FRAGMENT



HARDWARE AND EVENT CAPTURE

 Thin-film pressure mat CONFIGURE	Assembly start and part placement
 Workpiece camera VERIFY	Assembly state; exclude faces
 Arduino Nano serial test jig BUILD	Test and calibration pass/fail
 Reference scale VERIFY	Completed-node weight
 Approval button or UI CONFIGURE	Explicit deployment approval
 Raspberry Pi edge node ASSIGN	Evidence fusion, time, local buffer

START AT THIS TABLE

- 1 Verify the base, lid, kit, work order, size, and design revision identities.
- 2 Tare the pressure mat and reference scale.
- 3 Connect the workpiece camera and serial test jig. Load the approved procedure version.
- 4 Start capture before the first part enters the assembly area.
- 5 Assemble the node. Run the complete test and calibration procedure.
- 6 Approve only after a passing record. Use Resolve failure and Rework sensor after a failure.

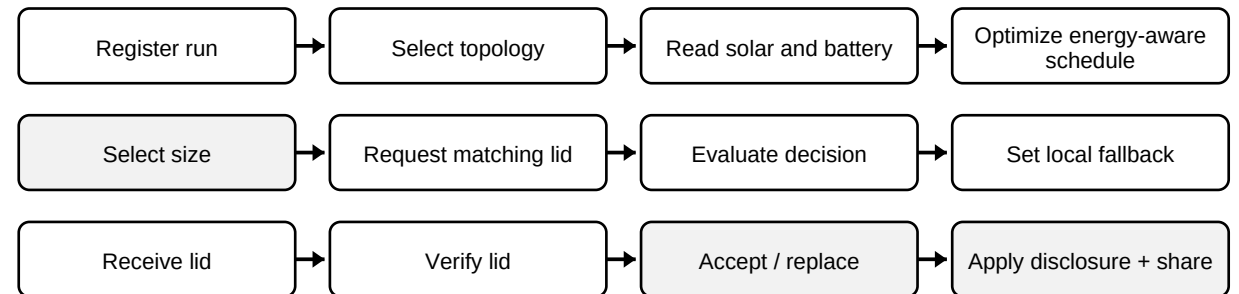
COMPLETE WHEN THIS EVIDENCE EXISTS

Product and component links, assembly start/end, test record, calibration record, rework history, final weight, and approval.

BPMN ACTIVITIES AT THIS TABLE

- Register run
- Select topology
- Read solar and battery
- Optimize energy-aware schedule
- Select standard container
- Select large container
- Request matching lid
- Evaluate decision
- Set local fallback
- Receive lid
- Verify lid
- Accept delivery
- Request replacement
- Apply disclosure
- Share outcome

PROCESS FRAGMENT



HARDWARE AND EVENT CAPTURE

 Management PC DOCUMENTED	Run, schedule, delivery, and outcome logs
 Inverter and BMS adapter NEEDED	Energy snapshot, unit, and freshness
 Network scale VERIFY	Lid received and removed
 XIAO workpiece camera VERIFY	Lid presence and size
 ID scanner or manual entry NEEDED	Shipment, lid, size, and revision identity
 MQTT broker and edge node DECIDE	Ingestion time, device state, local events

START AT THIS TABLE

- 1 Open the management interface. Verify broker connectivity and synchronized clocks.
- 2 Verify fresh inverter and battery readings with the documented units.
- 3 Tare the network scale. Test the camera region and identifier input.
- 4 Register the run and work order. Select the topology and due window.
- 5 Start capture. Create the schedule only from the current complete energy snapshot.
- 6 Receive and verify the lid. Share only the minimized outcome after Apply disclosure.

COMPLETE WHEN THIS EVIDENCE EXISTS

Run and work-order IDs, energy snapshot, schedule and policy versions, lid identity, delivery decision, disclosure decision, and outcome.